



Republic of Rwanda
Ministry of Education



LSVMS501

MINE SURVEYING

Perform Mine Surveying

Competence

RQF Level: 5

Learning Hours



Credits: 10

Sector: Construction and Building Services

Trade: Land Surveying

Module Type: Specific

Curriculum: CBSLSV5002-TVET certificate V in Land Surveying

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Purpose statement	The module describes the knowledge, skills and attitude required to perform mine surveying. At the end of this module, the learner should be able to plan the mine surveying work, collect mine surveying data, and produce mine plans and maps.					
Learning assumed to be in place	Apply land law and regulations, maintain surveying lab equipment, Perform cadastral surveying and Apply remote sensing technology.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment			50%

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Plan the work	1.1. Mining basics are properly described according to mining methods
	1.2. Mining legal and regulatory requirements are properly described according to their types
	1.3. Plan of work is properly prepared according to the specific task
2. Collect mine surveying data	2.1. Mine survey data is properly identified according to the types of work
	2.2. Surface mine surveying data is accurately collected based on selected surveying method
	2.3. Underground mine surveying data is accurately collected based on selected surveying method
3. Produce mine Plans and maps	3.1. Mining plans and maps are well identified according to their use
	3.2. Mine surveying data are accurately processed according to the types of product to be generated
	3.3. Mining plans and maps are accurately generated according to the work requirements

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Identifying of mine Surveying technology ✓ Identifying of mine surveying tools and equipment ✓ Identifying mining plan and map ✓ Describing mine Surveying principles ✓ Describing principles of ethics and mining regulatory compliance ✓ Describing GIS and Cartography principles ✓ Describing principles of safety, health and security ✓ Identifying work planning procedure	✓ Designing the work plan ✓ Applying environmental management skills ✓ Applying mine surveying computations ✓ Operating mine surveying tools and equipment ✓ Applying GIS skills in mine mapping ✓ Applying communication skills ✓ Use mine surveying CAD Software ✓ Performing mine surveying data Collection ✓ Applying ICT Skills	✓ Having critical thinking ✓ Having Team working spirit ✓ Having responsible ✓ Being Persistent ✓ Having Time management spirit ✓ Being Decisive ✓ Being Research oriented ✓ Being Innovative ✓ Being Flexible ✓ Being Attentive ✓ Being Detail-oriented ✓ Being Prompt ✓ Being Confident

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Plan the work 2. Collect mine surveying data 3. Produce mine Plans and maps
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Learning outcome 1: Plan the work	Learning hours: 30 hours
Indicative content	
• Describing mining basics ✓ Introducing mining  Mineral extraction  Mineral screening  Mine surveying ✓ Identifying mining methods  Surface mining	

- ✚ Underground mining
- ✚ Placer mining
- ✚ In-situ mining
- ✓ Describing mining procedures
 - ✚ Prospecting
 - ✚ Exploration
 - ✚ Extraction
 - ✚ Processing
 - ✚ Reclamation
- **Describing mining legal and regulatory requirements**
 - ✓ Mining law and regulations
 - ✚ Mineral licence requirements
 - ✚ Rights and responsibilities of the holder of a mineral licence
 - ✚ Land owner's rights
 - ✓ Environmental protection, health and safety
 - ✚ Health and safety in mining
 - ✚ Dynamites use and regulations
 - ✚ Environmental protection and rehabilitation
- **Preparing work-plan**
 - ✓ Identifying mine surveying activities
 - ✓ Selecting surveying resources
 - ✚ Equipment
 - ✚ Tools
 - ✚ Materials
 - ✚ Labours
 - ✓ Estimating time frame
 - ✓ Estimating the budget

Resources required for the indicative content

Equipment	Projector, computer, whiteboard, flipchart stand
Materials	Flip charts, papers, markers, duster,
Tools	Law on mining and quarry operations, Law governing land in Rwanda
Facilitation techniques	Presentation, Case study, simulation Practical exercises, individual/group works • The trainer presents to learners the basics of mine and used the videos to show mining methods and procedures and safety rules to be followed in mine surveying. • Trainer form small and manageable groups of learners and gives them a scenario simulating a mining surveying project and then asks them elaborate a work plan of that project

	- Trainer asks learners to present their work and then present the procedure to be followed when preparing the work plan. He/she may also present to them the well prepared work plan similar to the scenario provided - The trainer will ask the learners to use the skills gain through the presentation and then prepare a well-developed work plan. - Trainer monitor the work and provides clarification where necessary
Formative assessment methods	Written assessment, Oral assessment, Product based assessment.

Learning outcome 2: Collect mine surveying data		Learning hours: 40 hours
Indicative content		
Identifying mine survey data ✓ Mine site boundaries ✓ Mineral locations ✓ Utilities locations ✓ Topography ✓ Tailings ✓ Ores Collecting surface mine surveying data ✓ Setting up instruments ✓ Establishing control points ✓ Capturing surveying data Collecting underground mine surveying data ✓ Setting up instruments ✓ Transferring control points ✓ Recording surveying data  Tunnel alignment networks (nail heads or stakes)  Station marks along tunnel walls  Utilities		
Resources required for the indicative content		
Equipment	Projector, computer, whiteboard, flipchart stand, GNSS receivers, Total station, Laser scanners, UAVs, Levels, wheelbarrow	
Materials	Flip charts, papers, markers, duster, surveyor nails, peg, paint, cement, gravel, sand, warning tape	

Tools	Pick axe, Spade, club hammer, claw hammer, measuring tape, nylon building line , umbrella, reflecting jacket, safety boot, helmet, gloves, mask, headlight, ranging rod, communication radio, plumb bob, external disk
Facilitation techniques	Group discussions, Presentation, demonstration, trainer guide, field practice, Field visit • Trainer form small groups of learners and ask them to discuss on the types of data to be collected in mine surveying. • Trainer asks the learners to present their group findings and group the similar ideas from the groups work. • Trainers provide expert view on the types of mine surveying data. • Trainer presents to the learners the video showing how surveying data are collected for both surface and underground mine surveying • Trainer forms small groups of learners and ask them to go on the field and collect mine surveying data • Trainer guides and supports the learners during the process of data collection and ensures that the data are accurately collected • Trainer prepare field visit on the mine quarry and ask the learner to produce a report of what they gained from the visit.
Formative assessment methods	• Written assessment, • Oral assessment • Performance assessment • Product based assessment

Learning outcome 3: Produce mine Plans and maps	Learning hours: 30 hours
Indicative content	
• Identifying mining plans and maps ✓ Topographic maps ✓ Layout and detailed plans ✓ Profiles plans ✚ Longitudinal ✚ Cross sectional • Processing mine surveying data ✓ Download and upload data ✓ Visualising data ✓ Manipulating and analysing data • Generating mining plans and maps ✓ Designing plans and maps	

-  Topographic maps
-  Layout and detailed plans
-  Profiles plans

- ✓ Checking quality work
- ✓ Customizing layout
- ✓ Plotting and publishing mining plans and maps

Resources required for the indicative content

Equipment	Projector, computer, whiteboard, flipchart stand, plotter, printer, scanner
Materials	Flip charts, papers, markers, duster
Tools	AutoCAD civil 3D and ArcGIS Software, external disks,
Facilitation techniques	Group discussions, presentation, demonstration, trainer guide, field practice, think-pair-square-share. ● Trainer ask each learner to identify the types of mining plans and map, then asks two learners sited together to discuss on the same topic and the trainer monitor the activity ● Trainer asks two group to join together and form a group of four learners and ask them to discuss on the topic and write-down their findings and trainer monitors the activity ● Trainer asks the groups to present their findings and then after, the trainer provides his/her expert view. ● Trainer presents to the learners the video showing the process on how mine plans and maps are produced ● Trainer guides learner to process the data collected and then generate mine plans and maps.
Formative assessment methods	● Written assessment ● Oral assessment ● Performance assessment ● product based assessment

Integrated/Summative assessment

Integrated situation

In order to perform mining activities in Rwanda, interested companies have to submit all required documents at Rwanda Mines, Petroleum and Gas Board (RMB). The ABC Company Limited located in Rulindo district, Shyorongi sector, Bugaragara cell, Kabaraza village has signed a contract with you as Mine Surveyor to produce topographic map as one of the required documents.

Therefore, you are required to perform the following tasks in 10 hours:

-  Establish control points on the site
-  Demarcate mining site
-  Collect all surveying data related to natural and manmade features
-  Generate topographic map

Equipment, tools, materials and other logistics will be provided by the company.

Resources

Tools	Pick axe, Spade, machete, club hammer, claw hammer, measuring tape, umbrella, reflecting jacket, ranging rod, communication radio.
Equipment	GNSS Receiver, Total station and accessories, computer, printer, PPE
Materials/ Consumables	surveyor nails, peg, paint, cement, gravel, sand, warning tape

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: Plan the work 20%	1.1. Mining legal and regulatory requirements are well respected	Working place is safely managed			1
		Safely use of equipment and tools			2
		PPE are worn			2
	1.2. Work-Plan is properly prepared	Mine surveying activities are identified			5
		Surveying Instruments are selected			5
		Surveying tools are selected			3
		Surveying materials are selected			2
Learning outcome 2: Collect mine surveying data 40%	2.1. Mine survey data are properly collected	Mining site is delineated			10
		Control points are measured			5
		Control points are marked on the field			5
		Ground spot levels are captured			10
		Locations of man-made features are captured			5
		Locations of natural features are captured			5
Learning outcome 3: Produce	3.1. Mine surveying data are accurately	Mine surveying data are downloaded and uploaded into compatible format			2

mine Plans and maps 40%	processed	Surveying data are screened			3
		Displayed data are labeled			5
		Data analyzed			5
	3.2. Mining plans and maps are accurately produced	Contours are created			5
		Contours are labeled			3
		Man-made and natural features are mapped			7
		Map layout is customized			5
		Topographic map is plotted			5
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

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